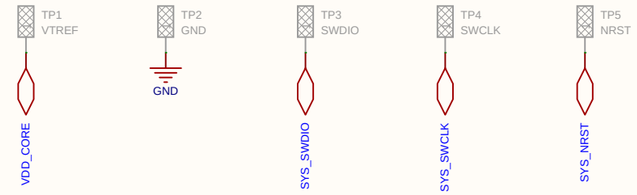
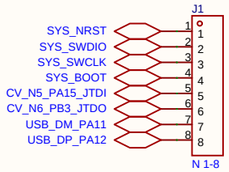
CraftVia Core-S G431 | v1 Rev0 - SCHEMATIC DRAFT

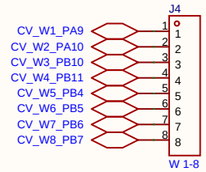
SWD RESCUE ACCESS / VTREF, GND, SWDIO, SWCLK, NRST



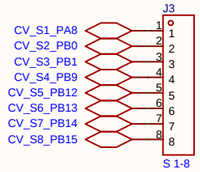
NORTH / J1.1-8 = N1-N8



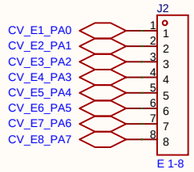
WEST / J4.1-8 = W1-W8



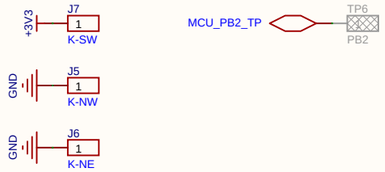
SOUTH / J3.1-8 = S1-S8



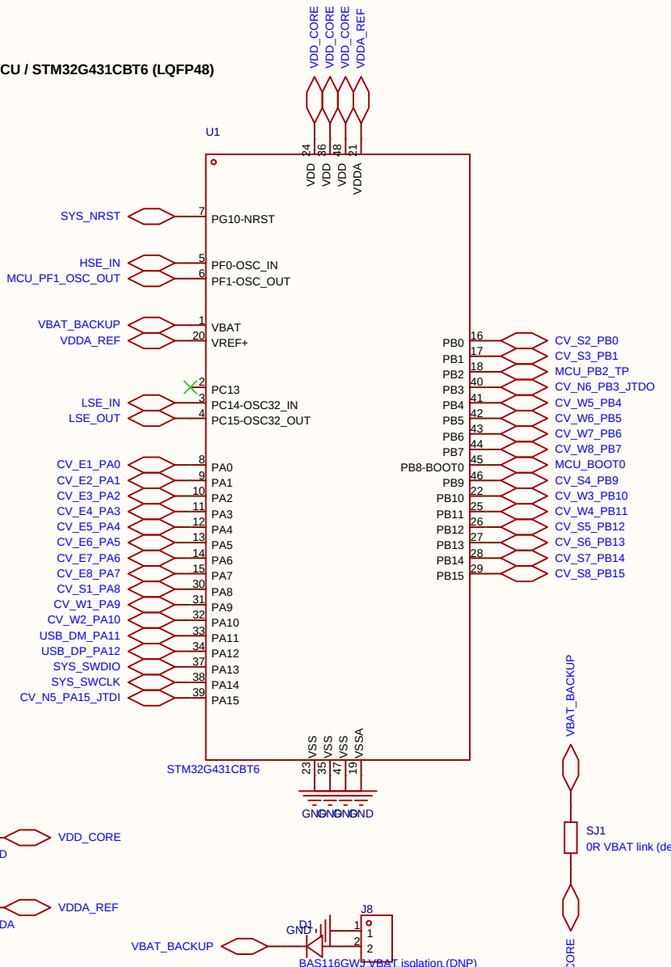
EAST / J2.1-8 = E1-E8



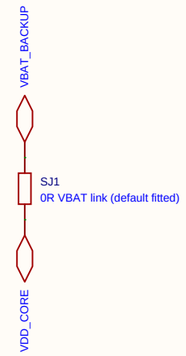
KEY / POWER



MCU / STM32G431CBT6 (LQFP48)







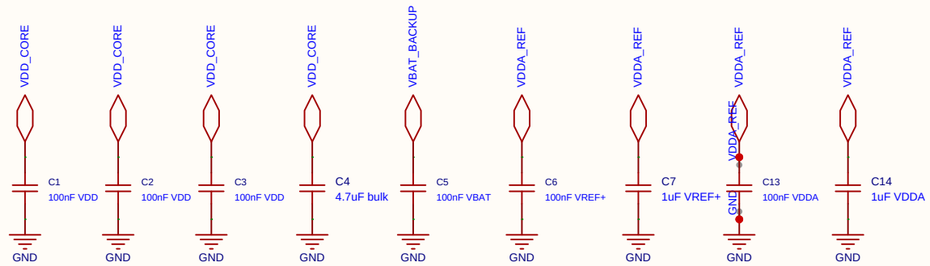
POWER / DECOUPLING

R1: +3V3 to VDD_CORE; R2: VDD_CORE to VDDA_REF; both 0R default.

C1-C3: VDD 100nF local. C4: bulk. C5: VBAT 100nF local.

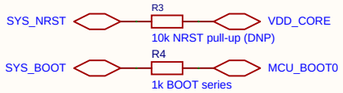
C6/C7: VREF+ 100nF + 1uF. C13/C14: VDDA 100nF + 1uF.

Place each analog pair at its MCU pin with the shortest return to VSSA/GND.



POWER FILTER / RESET OPTIONS

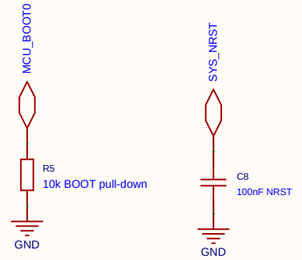
RESET / BOOT0



N1=PG10-NRST. N4 through R4=1k to PB8-BOOT0.

R5=10k BOOT0 pull-down. R3 NRST pull-up is DNP.

DECOUPLING / RESET



CLOCK SELECTION / ON-CORE OSCILLATOR OPTION

HSE: X1=8MHz, R6=0R start, C9/C10=18pF start; tune on PCB.

LSE: X2=32.768kHz, C11/C12=10pF start; tune and verify LSEDRV.

DESIGN NOTES / NOT FOR FABRICATION

3.3 V input. Internal HSI standard; no regulator / USB / LED.

Default clocks: HSI16 + PLL; HSI48 + CRS for USB; HSE/LSE DNP.

N2/N3: SWDIO/SWCLK. N5/N6: debug AUX. N7/N8: USB D-/D+.

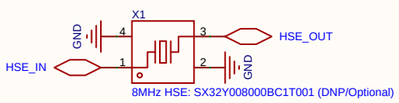
STM32G431 includes USB FS matching and the D+ pull-up.

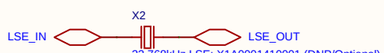
PC13 and PB9 are intentionally NC. USB/FDCAN on N7/N8 is exclusive.

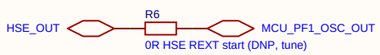
J1-J4: 2.54 mm headers; J5/J6=GND, J7=+3V3.

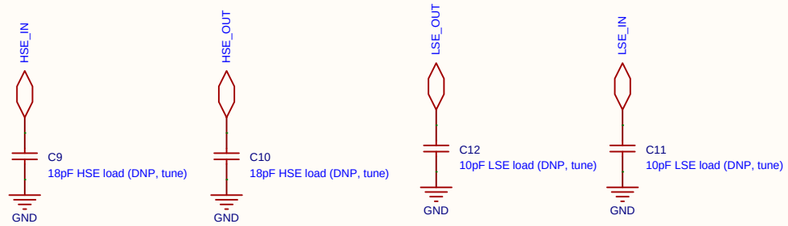
No regulator or reverse-polarity, overvoltage, overcurrent or hot-plug protection.

Common contact positions: see CraftVia Core-S Interface v1.









Schematic	Core-S-G431-v1-Rev0	Create at	2026-09-03
		Update at	2026-10-26
Board	Board1	Page	Core-S-G431
Drawn		CraftVia-Core-S-G431-v1-Rev0	
Reviewed			
		Version	Size / Page 1 Total 1
EasyEDA		V1 Rev0	A3 / CraftVia.org